Part-A → Screenshot_A4_PartA.JPG.

A1 → Block-1 -

Block-1 :

print(a.name) = Ravi
print(b.name = Priya

Block-2 :

Print("a starting up")

Output - starting up.

Block-3 :

Print(a.__balance)

Output - raise attribute Error.

Block-4 :

```
try:
    f = open("no_such_file.txt")
except FileNotFoundError:
    print("file missing")
finally:
    print("done")
```

File missing
done.

A2 → What does __init__ do and who calls it?

① It assigns values to the object's variable using self. and it returns nothing.

→ Instance of the class Calls it.

② What does self refer to?

→ Self is used as the first Parameter in a class method. It represents the specific object.

③ What is the difference between opening a file with "w" and with "a"?

→ If the file already exists, opening it with "w" mode removes all it previous data and if the file does not exist, a new file is created.

→ If the file already exists, opening it in "a" mode keeping all existing content intact. If the file does not exist, a new file is created.

④ Name one thing that can go wrong when reading a file, and the exception Python raises for it.

→ The file we are trying to read but it does not exist. and the exception Python raises a FileNotFoundError.

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B1→ Part-B- Student_A1-B.Py

```
class student:
    def __init__(self, name, marks):
        self.name = name
        self.__marks = 0
        self.__marks(marks)
    def set_marks(self, Value):
        if 0<= Value <= 100:
            self.__marks = Value
        else:
            Print("Error: Marks must be between 0 and 100")
    def get_marks(self):
        return self.__marks
    def grade(self):
        if self.__marks >= 80:
            return "A"
        elif self.__marks >= 60:
            return "B"
        elif self.__marks >= 40:
            return "C"
        else:
            return "Fail"

    def __str__(self):
        return f"Name: {self.name}, Marks: {self.__marks}
                grade: {self.grade}"
```

B→ My_name = "Student Kumar"
Print("Details of three Students ---")
st1 = student("my_name", 90)
st2 = student("Ram", 65)
st3 = student("Sita", 39)

Print (st1)
Print (st2)
Print (st3)
Print ()

B-3→

```
Print ("--- Providing Validator tasks ---")
Print (f" Current status before changing : {st1}")
st1.set_marks (150)
Print (f" Current status after changing : {st1}")
```

Part-C- Sudhen_A1_8.py

C-1→

```
# text_file:
from student import Student

text_file = "students.txt"
with open(text_file, 'w') as file:
    file.write(f"{st1.name, st1.get_marks(), st1.grade()}\n")
    file.write(f"{st2.name, st2.get_marks(), st2.grade()}\n")
    file.write(f"{st3.name, st3.get_marks(), st3.grade()}\n")

print(f"--- successfully created: {text_file} ---")
print("\n--- Reading lines from the text_file ---")
with open(text_file, 'r') as file:
    for line in file:
        print(line.strip())
```

C-2→

```
# CSV_file
from student import Student
import csv

with open(filename, 'w', newline='', encoding='utf-8') as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Marks"])
    writer.writerow([st1.name, st1.get_marks()])
    writer.writerow([st2.name, st2.get_marks()])
    writer.writerow([st3.name, st3.get_marks()])
print(f"--- successfully created {filename} ---")
print("\n--- Reading lines from the filename ---")
with open(filename, 'r', newline='', encoding='utf-8') as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

C-3→

```
# Json file.
import Json
from student import Student
json_file = "student.json"
st_data = [
    { "name": st1.name, "marks": st1.get_marks(),
      "grade": st1.grade() },
    { "name": st2.name, "marks": st2.get_marks(),
      "grade": st2.grade() },
    { "name": st3.name, "marks": st3.get_marks(),
      "grade": st3.grade() }
]

with open(json_file, 'w', newline='', encoding='UTF-8')
  as file:
    json_file.dump(st_data, file, indent=4)

print(f" -- Successfully Created {json_file} -- ")
print("\n -- Reading lines from the json_file -- ")
with open(json_file, 'r', newline='', encoding='UTF-8')
  as file:
  loaded_data = json.load(file)
    for item in loaded_data:
      print(item)
```

Part D - Sudheer_A4_D.py

D1 →

```
def read_marks_file(filename):
    try:
        with open(filename, 'w', newline='', encoding='UTF-8')
        as file:
            content = file.read().strip()
            marked = int(content)
            print(f"Successfully read mark: {marked}")
            return marked
    except FileNotFoundError:
        print(f"Error: the missing file: {filename}")
    except ValueError:
        print(f"Error: The content of the file named
            {filename} is not a number")
    finally:
        print("file operation completed")
```

D2 →

```
file_text = "Bad_Marks.txt"
file_valid = "Good_Marks.txt"
file_missing = "Does_Not_exist.txt"
with open(file_text, 'w', newline='', encoding='UTF-8')
as file:
    file.write("Hello")
with open(file_valid, 'w' newline='', encoding='UTF-8') as file:
    file.write("75")
if os.path.exists(file_missing):
    os.remove(file_missing)
```

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```
Print("--- Scenario - 1st : file does not exists ---")
Read_marks_file(file_missing)
Print("\n" + "-" * 60 + "\n")
Print("--- Scenario - 2nd : file contains 'Hello' ---")
Read_marks_file(file_text)
Print("--- Scenario - 3rd : file contains '75' ---")
Read_marks_file(file_valid)
Print("\n" + "-" * 60 + "\n")

for temp_file in [file_text, file_valid]:
    if os.path.exists('temp_file'):
        os.remove(temp_file)
```